Cornell notes Turner style
Instructions

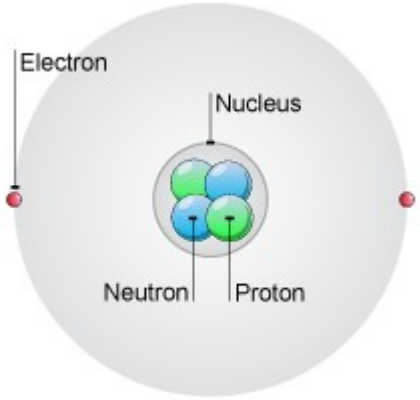
- Cut out the notes on the right hand side, and **paste** them into the **right hand side** of your notebook page. **Include** the **vertical black line or redraw this line** – giving yourself a large left hand margin
- **Read** the notes several times
- **Ask** yourself **three questions**:
  1. Can I recognise all the content (information) in the notes?
  2. In the notes is there a process or procedure I need to know?
  3. Do I understand the idea/meaning/argument outlined in the notes. That is: do I "get it"?
- **Do the following**
  - **a.** Within the notes themselves, **underline** all the **content** (information) you should learn or memorise.
  - **b.** In the left margin **label** each section of the notes as either a **definition**, **example**, **explanation**, **process**, or **evidence** (and you could make up your own categories)
  - **c.** With each label in step b, write a key word or phrase which tells us what the label is referring to. An example would be…

label → Definition
atom
Key
(note – the example is not for these notes
- The key words are highlighted)

  - **d.** At the end of the notes **write a summary underneath the notes**. The summary should be **short** – it is to **demonstrate your understanding** of the notes, not rehash the content. Often you should try to draw a diagram or mind map.

Atoms – Protons, Neutrons, and Electrons

All material things are made from atoms. An atom is the very smallest particle which exists of an element. All of the atoms of any one element (say oxygen) are identical. Oxygen gas is made from trillions of identical oxygen atoms. There are just over one hundred elements in the periodic table, so there are just over one hundred types of atoms in the universe. Luckily, atoms can join together in millions of different combinations to make **all** the substances on Earth and beyond.



Structure of the atom

Every atom is made of a **nucleus** consisting of **protons** and **neutrons**. The nucleus is surrounded by **electrons**.

It is the number of protons an atom has which gives it its identity, so for example, all oxygen atoms have exactly 8 protons.

Protons are positively charged and electrons are negatively charged. Neutrons have no charge. This means the nucleus (protons and neutrons) of an atom is positively charged. The negatively charged electrons move around the nucleus and are held in place by their attraction to the positively charged nucleus.

An atom has a **neutral** overall charge because it usually has the same number of electrons as protons (same number of negative and positive charges).

Protons and neutrons have the same **mass**. Electrons have such a small mass that this can usually be taken as zero.

Comparing the charge and mass of electrons, protons and neutrons

	Proton	Neutron	Electron
Charge	+1	0	-1
"Relative" Mass	1	1	0.0005 (almost zero)

The **atomic number** (also called the proton number) is the number of protons in an atom.

The **mass number** (also called the nucleon number) is the total number of protons and neutrons in an atom.

The elements are arranged in the periodic table in ascending order of atomic number so it's easy to find the name or symbol for an atom if you know the atomic number.

Elements are described by their atomic and their mass numbers, for example

Mass Number (p + n) → 23
Symbol of element → Na
Atomic number (p) → 11

Sodium (Na) which has 11 protons, 12 neutrons and 11 electrons